

Mathematics

A Cultural Heritage

Lecture 4

New Kinds of Math – Part 2: Counting Problems and Number Theory

Prime numbers

- Sieve of Eratosthenes (Greece, c. 276BC – 195BC):

	2	3	4	5	6	7	8	9	10	Prime numbers
11	12	13	14	15	16	17	18	19	20	
21	22	23	24	25	26	27	28	29	30	
31	32	33	34	35	36	37	38	39	40	
41	42	43	44	45	46	47	48	49	50	
51	52	53	54	55	56	57	58	59	60	
61	62	63	64	65	66	67	68	69	70	
71	72	73	74	75	76	77	78	79	80	
81	82	83	84	85	86	87	88	89	90	
91	92	93	94	95	96	97	98	99	100	
101	102	103	104	105	106	107	108	109	110	
111	112	113	114	115	116	117	118	119	120	



"Sieve of Eratosthenes animation". Licensed under CC BY-SA 3.0 via Wikimedia Commons

Number Theory: Number of Primes

- Infinitely Many Primes
- Prime Number Theorem
(Hadamard/Vallée-Poussin 1896):



Sir Michael Atiyah
1929-2019

$$\frac{\# \text{primes} \leq N}{N} \approx \frac{1}{\ln N}$$

- How good is this approximation? (\$1 Million USD)

Simple Questions, Complicated Answers

- Ramanujan (1887-1920):
- Contacted G. H. Hardy 1913:
Ramanujan's theorems "must be true, because, if they were not true, no one would have the imagination to invent them". – G. H. Hardy
- Partitions: 7 Sums equaling 5: $p(5)=7$

$$\begin{aligned}5 &= 4 + 1 = 3 + 2 = 3 + 1 + 1 \\ &= 2 + 2 + 1 = 2 + 1 + 1 + 1 \\ &= 1 + 1 + 1 + 1 + 1\end{aligned}$$



Partitions

- But $p(200)=\mathbf{3,972,999,029,388!}$
- How fast does it grow?
- Hardy, Ramanujan ($e \approx 2.1828$):

$$\# \text{ partitions of } n \approx \frac{1}{4n\sqrt{3}} e^{\pi\sqrt{\frac{2n}{3}}}$$

The Number 1729

“I remember once going to see him when he was ill at [Putney](#). I had ridden in taxi cab number 1729 and remarked that the number seemed to me rather a [dull one](#), and that I hoped it was not an unfavorable [omen](#). ”

– G. H. Hardy

“ ‘No,’ he replied, ‘it is a very interesting number; it is the smallest number expressible as the sum of two cubes in two different ways.’ ”



Fermat's Last Theorem

- Pythagorean triples:

$$x^2 + y^2 = z^2$$

- $x^3 + y^3 = z^3?$ $x^4 + y^4 = z^4?$...?

- Fermat (1637) claimed impossible:

Cubum autem in duos cubos, aut quadratoquadratum in duos quadratoquadratos, et generaliter nullam in infinitum ultra quadratum potestatem in duos eiusdem nominis fas est dividere cuius rei demonstrationem mirabilem sane detexi. Hanc marginis exiguitas non caperet.

It is impossible to separate a cube into two cubes, or a fourth power into two fourth powers, or in general, any power higher than the second, into two like powers. I have discovered a truly marvellous proof of this, which this margin is too narrow to contain.

- Wiles (1995): Fermat's claim true!

Basic Idea

- Moving problem is key:

$$x^n + y^n = z^n$$



$$Y^2 = X^3 + AX^2 + BX + C$$



???

Solution

(x, y, z, n)

Elliptic Curve

(A, B, C)

Modular Form

What are Modular Forms?

“Modular forms are functions on the complex plane that are inordinately symmetric. They satisfy so many internal symmetries that their mere existence seem like accidents. But they do exist.”

— Barry Mazur

